

Daniel Ayen

Full-Stack Software Engineer

Specialization: Highly Scalable Web Applications & Clean Architecture

PROFESSIONAL SUMMARY

Performance-driven Full-Stack Software Engineer with extensive experience designing, building, and deploying production-ready web applications. Deeply committed to writing clean, maintainable, and modular code by strictly adhering to software design principles like SOLID, DRY, and KISS, as well as Gang of Four (GoF) architectural patterns. Specialized in separating concerns cleanly between routes, controllers, and services to build highly testable backend systems. Adept at relational database design, normalization, and building seamless, animated user interfaces.

TECHNICAL SKILLS

Languages:	JavaScript (ES6+), TypeScript, Python, C++, Java, HTML5, CSS3
Frontend:	React, Next.js, Framer Motion (Advanced Animations), Bootstrap, Tailwind CSS
Backend & APIs:	Node.js, Express, RESTful API Design, MVC Architecture, Layered/Service-Layer Architecture
Databases:	MySQL, PostgreSQL (Relational Database Normalization, Complex Queries, Indexing)
Methodologies:	SOLID, DRY, KISS, Gang of Four (GoF) Design Patterns, Machine Learning (K-Means, KNN, Decision Trees)

CORE PROJECTS & EXPERIENCE

Enterprise-Grade Hospital Management System (HMS) — Lead Architect

Architecture: Node.js, Express, Service-Layer Pattern, PostgreSQL

- Designed a comprehensive, multi-module system tailored for clinical workflows, inventory management, and administrative tracking.
- Architected modular backend services to handle complex hospital roles, ensuring high data isolation and tight security constraints.

Student Allocation System — Independent Full-Stack Developer

Architecture: React, Node.js, Express, MySQL (Relational Normalization)

- Developed an end-to-end management system to automate freshman department selection and sectioning workflows.
- Engineered complex backend allocation logic based on student performance scores, specific preferences, and capacity constraints.
- Normalized a relational database structure to handle student records efficiently and prevent race conditions during selection.

AbeGarageProject (Garage Management System) — Lead Full-Stack Developer

Architecture: React, CSS, Bootstrap, Node.js, MySQL Connection Pooling

- Built and maintained a scalable garage management platform featuring modules for employee/customer tracking, service logs, and automated invoicing.
- Successfully troubleshooted and refactored critical database connection errors, transitioning the backend codebase to a stable connection pooling mechanism.

Personal Portfolio & Interactive Web Case Studies

Architecture: React, Framer Motion, Canva UI Customization

- Created a high-converting, professional multi-page responsive web portfolio utilizing React and Framer Motion for premium micro-interactions and animations.



EDUCATION & DEVELOPMENT

Bachelor of Science in Software Engineering (4th Year Student)

Injibara University

Introduction to AI

Coursera (Continuous Professional Development)